

AP[®] Physics C: Mechanics Practice Exam

From the 2015 Administration

This Practice Exam is provided by the College Board for AP Exam preparation. Teachers are permitted to download the materials and make copies to use with their students in a classroom setting only. To maintain the security of this exam, teachers should collect all materials after their administration and keep them in a secure location. **Exams may not be posted on school or personal websites, nor electronically redistributed for any reason.** Further distribution of these materials outside of the secure College Board site disadvantages teachers who rely on uncirculated questions for classroom testing. Any additional distribution is in violation of the College Board's copyright policies and may result in the termination of Practice Exam access for your school as well as the removal of access to other online services such as the AP Teacher Community and Online Score Reports.

ADVANCED PLACEMENT PHYSICS C TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS	
Proton mass, $m_p = 1.67 \times 10^{-27}$ kg Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg Electron mass, $m_e = 9.11 \times 10^{-31}$ kg Avogadro's number, $N_0 = 6.02 \times 10^{23}$ mol ⁻¹ Universal gas constant, $R = 8.31$ J/(mol·K) Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$ J/K	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C 1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19}$ J Speed of light, $c = 3.00 \times 10^8$ m/s Universal gravitational constant, $G = 6.67 \times 10^{-11}$ (N·m ²)/kg ² Acceleration due to gravity at Earth's surface, $g = 9.8$ m/s ²
1 unified atomic mass unit, Planck's constant, Vacuum permittivity, Coulomb's law constant, $k = 1/(4\pi\epsilon_0) = 9.0 \times 10^9$ (N·m ²)/C ² Vacuum permeability, Magnetic constant, $k' = \mu_0/(4\pi) = 1 \times 10^{-7}$ (T·m)/A 1 atmosphere pressure,	$1 \text{ u} = 1.66 \times 10^{-27}$ kg = 931 MeV/c ² $h = 6.63 \times 10^{-34}$ J·s = 4.14×10^{-15} eV·s $hc = 1.99 \times 10^{-25}$ J·m = 1.24×10^3 eV·nm $\epsilon_0 = 8.85 \times 10^{-12}$ C ² /(N·m ²) $\mu_0 = 4\pi \times 10^{-7}$ (T·m)/A $1 \text{ atm} = 1.0 \times 10^5$ N/m ² = 1.0×10^5 Pa

UNIT SYMBOLS	meter, m	mole, mol	watt, W	farad, F
	kilogram, kg	hertz, Hz	coulomb, C	tesla, T
	second, s	newton, N	volt, V	degree Celsius, °C
	ampere, A	pascal, Pa	ohm, Ω	electron volt, eV
	kelvin, K	joule, J	henry, H	

PREFIXES		
Factor	Prefix	Symbol
10 ⁹	giga	G
10 ⁶	mega	M
10 ³	kilo	k
10 ⁻²	centi	c
10 ⁻³	milli	m
10 ⁻⁶	micro	μ
10 ⁻⁹	nano	n
10 ⁻¹²	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following assumptions are used in this exam.

- I. The frame of reference of any problem is inertial unless otherwise stated.
- II. The direction of current is the direction in which positive charges would drift.
- III. The electric potential is zero at an infinite distance from an isolated point charge.
- IV. All batteries and meters are ideal unless otherwise stated.
- V. Edge effects for the electric field of a parallel plate capacitor are negligible unless otherwise stated.

ADVANCED PLACEMENT PHYSICS C EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \int \vec{F} dt = \Delta \vec{p}$$

$$\vec{p} = m\vec{v}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_N|$$

$$\Delta E = W = \int \vec{F} \cdot d\vec{r}$$

$$K = \frac{1}{2} m v^2$$

$$P = \frac{dE}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$\Delta U_g = mg \Delta h$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$I = \int r^2 dm = \sum m r^2$$

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

$$v = r \omega$$

$$\vec{L} = \vec{r} \times \vec{p} = I \vec{\omega}$$

$$K = \frac{1}{2} I \omega^2$$

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$a = \text{acceleration}$$

$$E = \text{energy}$$

$$F = \text{force}$$

$$f = \text{frequency}$$

$$h = \text{height}$$

$$I = \text{rotational inertia}$$

$$J = \text{impulse}$$

$$K = \text{kinetic energy}$$

$$k = \text{spring constant}$$

$$\ell = \text{length}$$

$$L = \text{angular momentum}$$

$$m = \text{mass}$$

$$P = \text{power}$$

$$p = \text{momentum}$$

$$r = \text{radius or distance}$$

$$T = \text{period}$$

$$t = \text{time}$$

$$U = \text{potential energy}$$

$$v = \text{velocity or speed}$$

$$W = \text{work done on a system}$$

$$x = \text{position}$$

$$\mu = \text{coefficient of friction}$$

$$\theta = \text{angle}$$

$$\tau = \text{torque}$$

$$\omega = \text{angular speed}$$

$$\alpha = \text{angular acceleration}$$

$$\phi = \text{phase angle}$$

$$\vec{F}_s = -k \Delta \vec{x}$$

$$U_s = \frac{1}{2} k (\Delta x)^2$$

$$x = x_{\max} \cos(\omega t + \phi)$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_G| = \frac{G m_1 m_2}{r^2}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY AND MAGNETISM

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \left| \frac{q_1 q_2}{r^2} \right|$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E_x = -\frac{dV}{dx}$$

$$\Delta V = -\int \vec{E} \cdot d\vec{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

$$U_E = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\Delta V = \frac{Q}{C}$$

$$C = \frac{\kappa \epsilon_0 A}{d}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$I = \frac{dQ}{dt}$$

$$U_C = \frac{1}{2} Q \Delta V = \frac{1}{2} C (\Delta V)^2$$

$$R = \frac{\rho \ell}{A}$$

$$\vec{E} = \rho \vec{J}$$

$$I = Ne v_d A$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$P = I \Delta V$$

$$A = \text{area}$$

$$B = \text{magnetic field}$$

$$C = \text{capacitance}$$

$$d = \text{distance}$$

$$E = \text{electric field}$$

$$\mathcal{E} = \text{emf}$$

$$F = \text{force}$$

$$I = \text{current}$$

$$J = \text{current density}$$

$$L = \text{inductance}$$

$$\ell = \text{length}$$

$$n = \text{number of loops of wire per unit length}$$

$$N = \text{number of charge carriers per unit volume}$$

$$P = \text{power}$$

$$Q = \text{charge}$$

$$q = \text{point charge}$$

$$R = \text{resistance}$$

$$r = \text{radius or distance}$$

$$t = \text{time}$$

$$U = \text{potential or stored energy}$$

$$V = \text{electric potential}$$

$$v = \text{velocity or speed}$$

$$\rho = \text{resistivity}$$

$$\Phi = \text{flux}$$

$$\kappa = \text{dielectric constant}$$

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \hat{r}}{r^2}$$

$$\vec{F} = \int I d\vec{\ell} \times \vec{B}$$

$$B_s = \mu_0 n I$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$U_L = \frac{1}{2} L I^2$$

ADVANCED PLACEMENT PHYSICS C EQUATIONS

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

$$s = r\theta$$

Rectangular Solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

A = area

C = circumference

V = volume

S = surface area

b = base

h = height

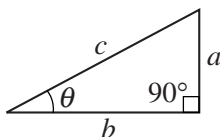
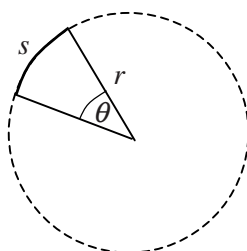
ℓ = length

w = width

r = radius

s = arc length

θ = angle



CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\ln ax) = \frac{1}{x}$$

$$\frac{d}{dx}[\sin(ax)] = a \cos(ax)$$

$$\frac{d}{dx}[\cos(ax)] = -a \sin(ax)$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \frac{dx}{x+a} = \ln|x+a|$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax)$$

VECTOR PRODUCTS

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

PHYSICS C: MECHANICS

SECTION I

Time—45 minutes

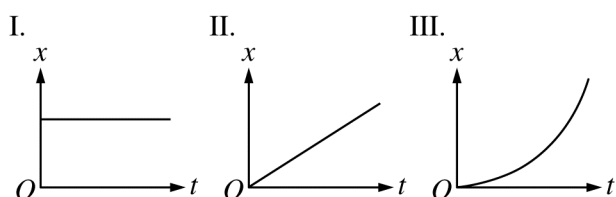
35 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

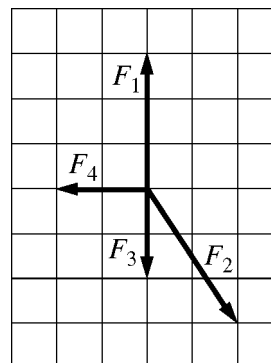
Note: To simplify calculations, you may use $g = 10\text{ m/s}^2$ in all problems.

Questions 1-3

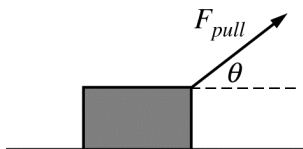
The graphs below show position x plotted as a function of time t for each of three objects constrained to move along a straight, level path.



- In which of the graphs is the magnitude of the momentum of the object increasing?
 - II only
 - III only
 - I and II
 - I and III
 - II and III
- In which of the graphs does the object have a constant nonzero kinetic energy?
 - I only
 - II only
 - III only
 - I and II
 - I and III
- In which of the graphs is the net force on the object zero?
 - I only
 - III only
 - I and II only
 - II and III only
 - I, II, and III



- The coplanar forces F_1 , F_2 , F_3 , and F_4 represented above can act on an object. If the object is initially at rest, which combination of the forces would result in static equilibrium?
 - F_1 and F_3 only
 - F_1 , F_2 , and F_4 only
 - F_2 , F_3 , and F_4 only
 - All four forces
 - No combination of the forces would result in equilibrium.



5. A block of mass M is being pulled to the right at a constant speed with a pull force F_{pull} that is at an angle of θ with respect to the horizontal, as shown in the figure above. The force of friction acting on the block has magnitude f . What is the magnitude of F_{pull} ?

(A) $f \cos \theta$
 (B) $f \sin \theta$
 (C) f
 (D) $f / \cos \theta$
 (E) $f / \sin \theta$

Questions 6-8

An astronaut hits a golf ball of mass m on the Moon, where there is no atmosphere and the acceleration due to gravity is $g/6$, where g is the acceleration due to gravity on Earth. Assume that the golf club is in contact with the ball for a time t . Just after losing contact with the club, the ball has an initial speed v directed at an angle θ above the horizontal.

6. What is the magnitude of the average force exerted by the club on the ball during contact?

(A) mv
 (B) $mv \cos \theta$
 (C) $\frac{mv}{t}$
 (D) $\frac{mg}{6}$
 (E) $\frac{mg}{6} \cos \theta$

7. What is the work done by the club on the ball?

(A) $\frac{1}{2}mv^2$
 (B) $\frac{1}{2}m(v \sin \theta)^2$
 (C) $\frac{1}{2}m(v \cos \theta)^2$
 (D) $6\left(\frac{1}{2}mv^2\right)$
 (E) $\left(\frac{1}{6}\right)\left(\frac{1}{2}mv^2\right)$

8. What is the maximum height above the Moon's surface attained by the ball?

(A) vt
 (B) $6vt$
 (C) $\frac{v^2}{2g}$
 (D) $\frac{(v \sin \theta)^2}{2g}$
 (E) $6\left[\frac{(v \sin \theta)^2}{2g}\right]$

9. A simple pendulum of length L has a period T . What is the length of another simple pendulum of period $2T$, assuming that both pendulums swing with small amplitude?

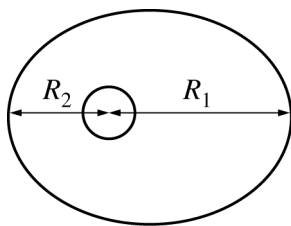
(A) $\frac{1}{2}L$
 (B) L
 (C) $\sqrt{2}L$
 (D) $2L$
 (E) $4L$

10. Pendulum 1 is displaced by a very small angle θ_1 , is released from rest, and oscillates with a period T_1 . Pendulum 2 is identical to pendulum 1, but it is displaced by a very small angle θ_2 , is released from rest, and oscillates with a period T_2 , where $\theta_1 > \theta_2$. Using the small angle approximation, how do the periods T_1 and T_2 compare?

- (A) $T_1 > T_2$
 (B) $T_1 = T_2$
 (C) $T_1 < T_2$
 (D) It cannot be determined without knowing the values of θ_1 and θ_2 .
 (E) It cannot be determined without knowing the length of the pendulums.

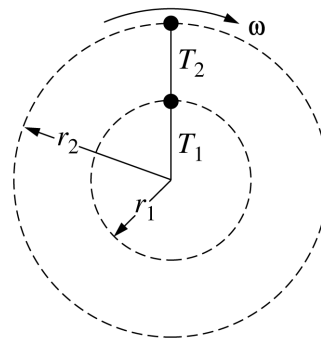
11. A newly discovered planet has three times the mass and five times the radius of Earth. What is the ratio of the acceleration due to gravity at the surface of the new planet to the acceleration due to gravity at the surface of Earth?

- (A) $3/25$
 (B) $3/5$
 (C) $5/9$
 (D) $9/5$
 (E) $25/3$



12. A satellite is in an elliptical orbit about a planet, as shown in the figure above. At apogee, a distance R_1 from the planet, the satellite's angular speed is ω . What is the angular speed of the satellite at perigee, a distance $R_2 = R_1/2$ from the planet?

- (A) $(1/4)\omega$
 (B) $(1/2)\omega$
 (C) $(1/\sqrt{2})\omega$
 (D) 2ω
 (E) 4ω

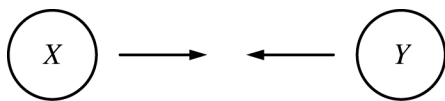


13. Two objects of equal mass m are tied to a string and placed on a horizontal frictionless tabletop. The string and objects spin with angular velocity ω so that the objects move in circles of radius r_1 and r_2 , as shown above. The tensions T_1 and T_2 must satisfy which of the following relations?

- (A) $T_1 - T_2 = m\omega^2 r_1$
 (B) $T_2 - T_1 = m\omega^2 r_1$
 (C) $T_1 + T_2 = m\omega^2 r_2$
 (D) $T_1 + T_2 = m\omega^2 r_1$
 (E) $T_1 = T_2$

14. A block of mass 3 kg is swung on a cord in a horizontal circle of radius 2 m. The speed of the block is 6 m/s. The magnitude of the acceleration of the block is most nearly

- (A) zero
 (B) 10 m/s^2
 (C) 18 m/s^2
 (D) 36 m/s^2
 (E) 54 m/s^2

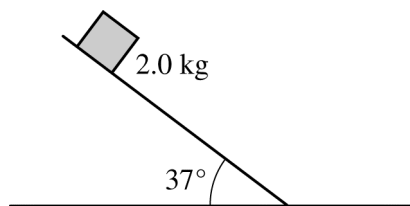


15. Two 1.0 kg balls have a perfectly elastic head-on collision, as shown in the figure above. Before the collision, ball X has a velocity of 2.0 m/s to the right and ball Y has a velocity of 4.0 m/s to the left. After the collision, ball X has a velocity of

(A) 0 m/s
 (B) 2.0 m/s to the right
 (C) 2.0 m/s to the left
 (D) 4.0 m/s to the right
 (E) 4.0 m/s to the left

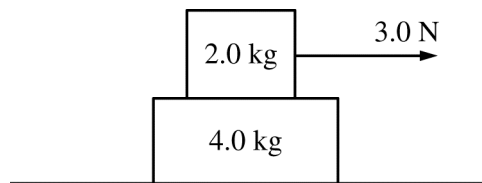
16. An object of mass m and velocity $3v$ toward the east has a completely inelastic collision with an object of mass $2m$ and velocity $2v$ toward the north. After the collision, the momentum of the combined object has a magnitude of

(A) $5mv$
 (B) $7mv$
 (C) $10mv$
 (D) $12mv$
 (E) $15mv$



17. A 2.0 kg block is released from rest at the top of a rough plane inclined at 37° to the horizontal, as shown above. The block slides down the incline with an acceleration of 4.0 m/s^2 . What is the approximate magnitude of the force of friction on the block as it slides down the incline?

(A) 4.0 N
 (B) 6.0 N
 (C) 10 N
 (D) 12 N
 (E) 20 N



18. A 2.0 kg box is on top of a 4.0 kg box that is placed on a frictionless surface, as shown in the figure above. A 3.0 N force is applied to the top box. What is the minimum coefficient of friction between the two boxes needed to keep them moving together?

(A) 0.10
 (B) 0.15
 (C) 0.30
 (D) 0.50
 (E) 1.0

Questions 19-20

A juggler has three objects in the air simultaneously. The masses of the objects are m , $2m$, and $3m$. At a particular instant, their respective vertical velocities are $\vec{v}$, 0, and $-\vec{v}$, where positive velocity is upward.

19. What is the magnitude of the acceleration of the center of mass of the three-object system at this instant?

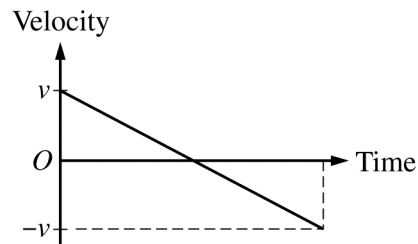
(A) zero
 (B) $g/6$
 (C) $g/3$
 (D) $g/2$
 (E) g

20. What is the velocity of the center of mass of the three-object system at this instant?

(A) $-\vec{v}/2$
 (B) $-\vec{v}/3$
 (C) zero
 (D) $\vec{v}/3$
 (E) $\vec{v}/2$

21. In a classroom demonstration, a teacher has a coin and feather in a long tube. The teacher uses a vacuum pump to remove all the air from the tube. The feather and coin are then dropped from the top of the tube at the same time. Which of the following describes what happens and why?

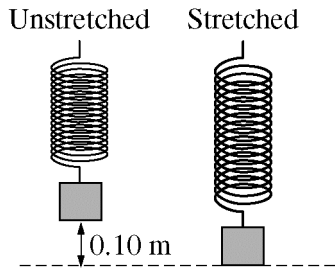
- (A) The coin hits the bottom of the tube first because it weighs more.
- (B) The coin hits the bottom of the tube first because it is more dense.
- (C) The feather hits the bottom of the tube first because it weighs less.
- (D) The coin and feather hit the bottom of the tube at the same time because they weigh the same.
- (E) The coin and feather hit the bottom of the tube at the same time because they have the same acceleration.



22. The graph above shows velocity versus time for an object moving in a straight line, initially in the $+x$ direction. Which of the following is true for the motion of the object for the entire time shown on the graph?

- (A) Its speed decreases with constant acceleration.
- (B) Its speed increases with constant acceleration.
- (C) Its speed decreases with acceleration that also decreases in magnitude.
- (D) Its speed decreases to zero, then increases as the object moves in the $-x$ direction.
- (E) Its acceleration decreases at a constant rate to zero, then increases at a constant rate.

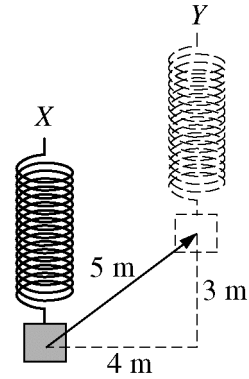
Questions 23-24



Note: Figure not drawn to scale.

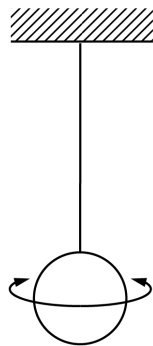
A block is hung vertically from a spring of force constant 50 N/m and slowly lowered to equilibrium, stretching the spring by 0.10 m , as shown in the figure above.

23. If instead of being lowered slowly to equilibrium, the block—still attached to the spring—was released from rest and allowed to fall, which of the following would happen?
- (A) It would descend 0.10 m and remain at that position.
 - (B) It would descend 0.10 m and then reverse its direction of motion.
 - (C) It would descend 0.20 m and remain at that position.
 - (D) It would descend 0.20 m and then reverse its direction of motion.
 - (E) It would descend more than 0.20 m and then reverse its direction of motion.



Note: Figure not drawn to scale.

24. The spring and attached block are allowed to come to rest at position X and are then moved to rest at position Y along the diagonal path shown above. What is the work done on the block by the spring in moving it along this path?
- (A) 0 J
 - (B) 15 J
 - (C) 20 J
 - (D) 25 J
 - (E) It cannot be determined without being given the value of the mass of the block.



25. The sphere shown above, which hangs from a rigid wire, is given a twist and set into angular harmonic motion. The resulting equation of motion of the oscillating sphere is $\theta(t) = 0.1 \cos(2t)$, where θ is in radians and t is in seconds. If the rotational inertia of the sphere is $10 \text{ kg}\cdot\text{m}^2$, then the torsion constant of the wire is most nearly

- (A) $7 \text{ kg}\cdot\text{m}^2/\text{s}^2$
- (B) $20 \text{ kg}\cdot\text{m}^2/\text{s}^2$
- (C) $30 \text{ kg}\cdot\text{m}^2/\text{s}^2$
- (D) $40 \text{ kg}\cdot\text{m}^2/\text{s}^2$
- (E) $50 \text{ kg}\cdot\text{m}^2/\text{s}^2$

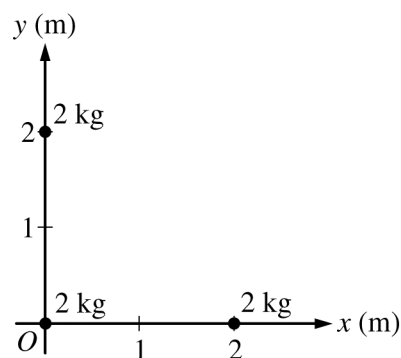
26. Assuming air resistance is negligible, how does a diver doing flips off a diving board change rotational speed in midair?

- (A) By changing rotational inertia
- (B) By maintaining a constant linear speed
- (C) By changing linear momentum
- (D) By exerting a net torque
- (E) By changing angular momentum

27. A toy spacecraft is launched directly upward. When the toy reaches its highest point, a spring is released and the toy splits into two parts with masses of 0.01 kg and 0.09 kg , respectively. Immediately after the separation, the 0.01 kg part moves horizontally due east. Air resistance is negligible. True statements about the 0.09 kg part include which of the following?

- I. It could move north immediately after the spring is released.
- II. It takes longer to reach the ground than does the 0.01 kg part.
- III. It strikes the ground closer to the launch point than does the 0.01 kg part.

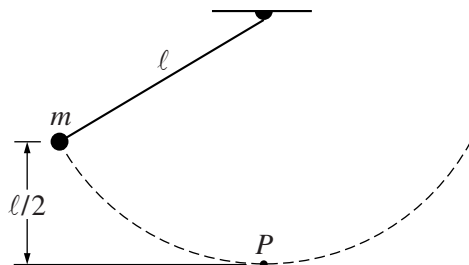
- (A) I only
- (B) III only
- (C) I and II only
- (D) II and III only
- (E) I, II, and III



28. A system consists of three 2 kg objects located at coordinates $(0, 2)$, $(0, 0)$, and $(2, 0)$, as shown in the figure above. What are the coordinates (x, y) of the center of mass of the system?

- (A) $(1/2 \text{ m}, 1/2 \text{ m})$
- (B) $(2/3 \text{ m}, 2/3 \text{ m})$
- (C) $(3/4 \text{ m}, 3/4 \text{ m})$
- (D) $(1 \text{ m}, 1 \text{ m})$
- (E) $(4/3 \text{ m}, 4/3 \text{ m})$

Questions 29-30 refer to the following.



A ball of mass m , suspended by a cord of length ℓ , is displaced along its arc until it is elevated a vertical distance of $\ell/2$ above its lowest position at point P , as shown above. The ball is then released from rest and swings like a simple pendulum.

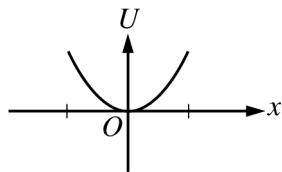
29. The speed of the ball as it swings through point P is

- (A) $\sqrt{g\ell}$
- (B) $\sqrt{2g\ell}$
- (C) $\sqrt{mg\ell}$
- (D) $\frac{mg\ell}{2}$
- (E) $mg\ell$

30. If air resistance and friction in the pivot are negligible, quantities that are conserved as the ball swings back and forth include which of the following?

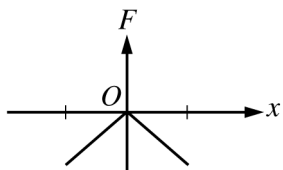
- I. Mechanical energy of the ball-Earth system
- II. Linear momentum of the ball
- III. Angular momentum of the ball about the pivot point

- (A) I only
- (B) II only
- (C) I and III only
- (D) II and III only
- (E) I, II, and III

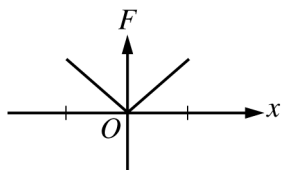


31. The graph above shows potential energy U versus position x for an object. Which of the following graphs shows the force F exerted on the object as a function of position x ?

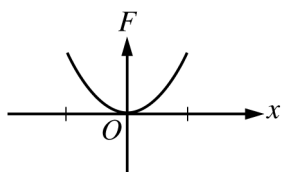
(A)



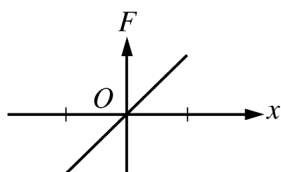
(B)



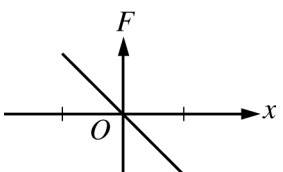
(C)

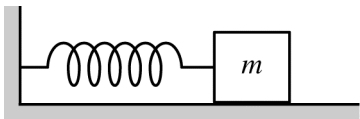


(D)



(E)





32. A block of mass m on a horizontal surface is attached to one end of a horizontal spring. The other end of the spring is fixed, as shown in the figure above. The block oscillates with negligible friction and an amplitude of 4.0 cm. How far from the equilibrium position is the block when the kinetic energy of the block is three times the potential energy of the system?

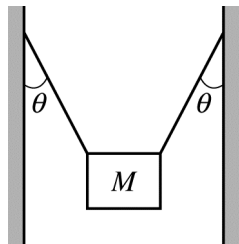
(A) 1 cm
 (B) $\sqrt{2}$ cm
 (C) 2 cm
 (D) $2\sqrt{2}$ cm
 (E) 3 cm

33. An Earth satellite in a circular orbit of radius R has a period T . What is the period of an Earth satellite in a circular orbit of radius $4R$?

(A) T
 (B) $2T$
 (C) $4T$
 (D) $6T$
 (E) $8T$

34. The net force on an object initially moving along the positive x -axis is given by $F(t) = (4t - t^3)$ for $t > 0$, where F is in newtons and along the x -axis, and t is in seconds. At what time does the object reach its maximum positive velocity?

(A) $\frac{3}{4}$ s
 (B) $\frac{4}{3}$ s
 (C) 2 s
 (D) $\sqrt{6}$ s
 (E) 3 s



35. A heavy sign of mass M is hung between two buildings from two supporting wires, each making an angle θ with the vertical, as shown in the figure above. What is the tension in each wire?

- (A) $\frac{Mg}{2\sin\theta}$
(B) $\frac{Mg}{2\cos\theta}$
(C) $\frac{Mg}{\sin\theta}$
(D) $\frac{2Mg}{\sin\theta}$
(E) $\frac{2Mg}{\cos\theta}$

STOP

END OF MECHANICS SECTION I

**IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON MECHANICS SECTION I ONLY.**

DO NOT TURN TO ANY OTHER TEST MATERIALS.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
- **TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET**

PHYSICS C: MECHANICS

SECTION II

Time—45 minutes

3 Questions

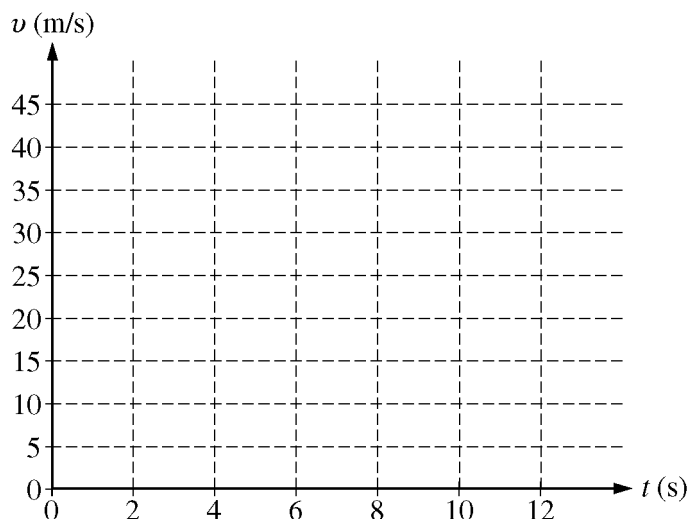
Directions: Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.

Mech. 1.

A new sports car is undergoing acceleration tests to determine its specifications. The following data on speed v versus time t are recorded for the car as it accelerates from rest along a straight track.

v (m/s)	0	14	23	30	35	39
t (s)	0	2	4	6	8	10

- (a) On the axes below, plot v as a function of t and sketch a curve that best represents the data.



- (b) Answer each of the following for the time period $t = 0$ s to $t = 10$ s.

i. Does the speed of the car increase, decrease, or stay the same?

____ Increase ____ Decrease ____ Stay the same

Justify your answer.

ii. Does the acceleration of the car increase, decrease, or stay the same?

____ Increase ____ Decrease ____ Stay the same

Justify your answer.

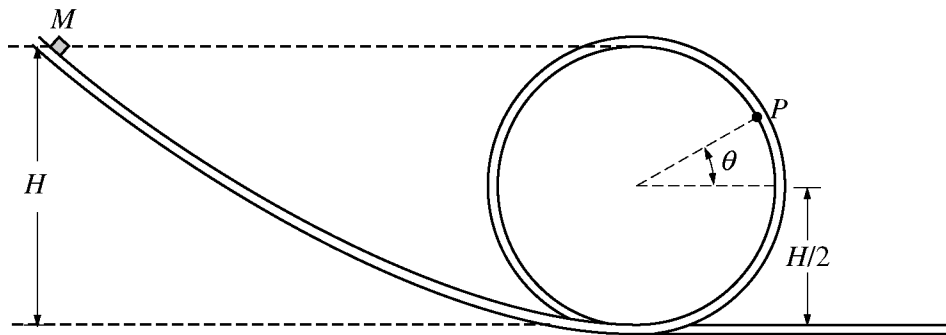
- (c) Explain how you would use your graph in part (a) to find the distance traveled by the car between $t = 2$ s and $t = 8$ s.

The equation for the speed v of the car as a function of time t found from the graph is $v(t) = -0.3t^2 + 7t$, where v is in meters per second and t is in seconds.

- (d) Derive an expression for the acceleration of the car $a(t)$ as a function of time t .

- (e) Calculate the position of the car as a function of time, $x(t)$, assuming that the car starts from rest at the origin of a coordinate system.

- (f) Calculate the distance traveled by the car between $t = 2$ s and $t = 8$ s.

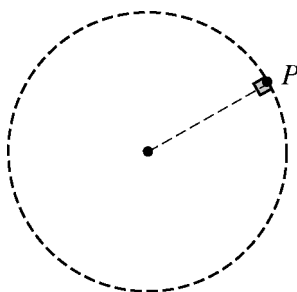


Mech. 2.

The model of a roller coaster shown above consists of a ramp and a circular loop. A small block of mass M starts from height H with speed $v = 0$ and slides without friction down the ramp and around the inside of the loop of radius $H/2$. Position P is a point on the loop at an angle θ with the horizontal, as shown above. Express all algebraic answers in terms of M , H , θ , and fundamental constants, as appropriate.

(a) Derive an expression for the speed of the block at position P .

(b) On the following diagram, draw and label vectors to represent the forces (not components) on the block when it is at position P . Each force must be represented by a distinct arrow starting on, and pointing away from, the dot at position P .



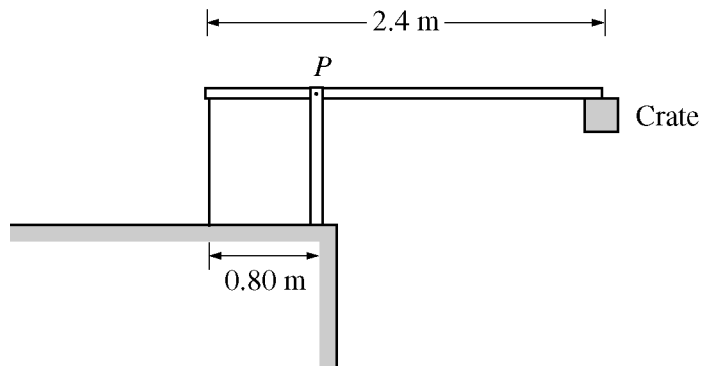
- (c) Derive an expression for the magnitude of the force exerted by the loop on the block when it is at position P . If you need to draw anything other than what you have shown in part (b) to assist in your solution, use the space below. Do NOT add anything to the figure in part (b).

- (d) Calculate the angle θ_L , in degrees, at which the block leaves the loop.

- (e) The small block is replaced with a small sphere that also starts at height H . Now there is enough friction to allow the sphere to roll down the ramp without slipping. Identify how the new angle θ_n at which the sphere leaves the loop compares with the angle θ_L calculated in part (d).

_____ $\theta_n < \theta_L$ _____ $\theta_n = \theta_L$ _____ $\theta_n > \theta_L$

Explain your reasoning.



Mech. 3.

A uniform, rigid, thin board that is 2.4 m long and weighs 200 N is attached to a post by a pivot at point P and hangs over the edge of a building, as shown above. A crate weighing 400 N is attached to the right end of the board. The left end of the board, 0.80 m from the pivot, is attached to a light vertical wire anchored at the other end to keep the board from rotating.

- (a) On the diagram below, draw and label vectors to represent all the forces (not components) acting on the board. Show each force vector originating at its point of application.



- (b) Calculate the magnitudes of the forces exerted on the board by the post and by the wire. If you need to draw anything other than what you have shown in part (a) to assist in your solution, use the space below. Do NOT add anything to the figure in part (a).

(c) The rotational inertia of the board alone about its center is $\frac{1}{12}ML^2$, where M is the mass of the board and L is its length. Calculate the rotational inertia of the combined board-crate system about point P .

(d) Suppose that the wire breaks and the board begins to pivot about point P . Calculate each of the following.

i. The magnitude of the initial angular acceleration of the board-crate system

ii. The magnitude of the initial linear acceleration of the left end of the board

**Answer Key for AP Physics C: Mechanics
Practice Exam, Section I**

Question 1: B	Question 19: E
Question 2: B	Question 20: B
Question 3: C	Question 21: E
Question 4: B	Question 22: D
Question 5: D	Question 23: D
Question 6: C	Question 24: B
Question 7: A	Question 25: D
Question 8: E	Question 26: A
Question 9: E	Question 27: B
Question 10: B	Question 28: B
Question 11: A	Question 29: A
Question 12: E	Question 30: A
Question 13: A	Question 31: E
Question 14: C	Question 32: C
Question 15: E	Question 33: E
Question 16: A	Question 34: C
Question 17: A	Question 35: B
Question 18: A	

AP[®] PHYSICS

2015 SCORING GUIDELINES

GENERAL NOTES ABOUT 2015 PHYSICS SCORING GUIDELINES

1. The solutions contain the most common method of solving the free-response questions and the allocation of points for this solution. Some also contain a common alternate solution. Other methods of solution also receive appropriate credit for correct work.
2. Generally, double penalty for errors is avoided. For example, if an incorrect answer to part (a) is correctly substituted into an otherwise correct solution to part (b), full credit will usually be awarded. One exception to this may be cases when the numerical answer to a later part should be easily recognized as wrong, e.g., a speed faster than the speed of light in vacuum.
3. Implicit statements of concepts normally receive credit. For example, if use of the equation expressing a particular concept is worth one point, and a student's solution contains the application of that equation to the problem but the student does not write the basic equation, the point is still awarded. However, when students are asked to derive an expression it is normally expected that they will begin by writing one or more fundamental equations, such as those given on the exam equation sheet. For a description of the use of such terms as “derive” and “calculate” on the exams, and what is expected for each, see “The Free-Response Sections—Student Presentation” in the *AP Physics; Physics C: Mechanics, Physics C: Electricity and Magnetism Course Description* or “Terms Defined” in the *AP Physics 1: Algebra-Based and AP Physics 2: Algebra-Based Course and Exam Description*.
4. The scoring guidelines typically show numerical results using the value $g = 9.8 \text{ m/s}^2$, but use of 10 m/s^2 is of course also acceptable. Solutions usually show numerical answers using both values when they are significantly different.
5. Strict rules regarding significant digits are usually not applied to numerical answers. However, in some cases answers containing too many digits may be penalized. In general, two to four significant digits are acceptable. Numerical answers that differ from the published answer due to differences in rounding throughout the question typically receive full credit. Exceptions to these guidelines usually occur when rounding makes a difference in obtaining a reasonable answer. For example, suppose a solution requires subtracting two numbers that should have five significant figures and that differ starting with the fourth digit (e.g., 20.295 and 20.278). Rounding to three digits will lose the accuracy required to determine the difference in the numbers, and some credit may be lost.

AP[®] PHYSICS C: MECHANICS

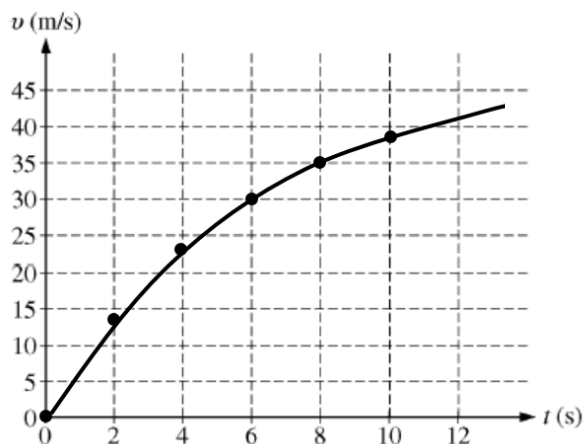
2015 SCORING GUIDELINES

Question 1

15 points total

**Distribution
of points**

(a) 3 points



For accurately plotting at least five points
 For a smooth curve with a positive slope
 For a concave down curve starting at the origin

1 point
 1 point
 1 point

(b)
 i. 1 point

The correct choice is “Increase”
 For a correct justification
 Example: The speed increases as the magnitude of the velocity increases for each data point

1 point

ii. 1 point

The correct choice is “Decrease”
 For a correct justification
 Example: Since the slope of the line decreases with time, the acceleration of the car must decrease with time

1 point

(c) 2 points

For mentioning area under the curve
 For specifically describing the area between $t = 2$ s and $t = 8$ s
 Example: To determine the distance traveled from a velocity-time graph, one should find the area under the curve. For time $t = 2$ s to $t = 8$ s, one can approximate the area using a trapezoid that approximately fills the area under the curve.

1 point
 1 point

AP[®] PHYSICS C: MECHANICS
2015 SCORING GUIDELINES

Question 1 (continued)

	Distribution of points
(d) 2 points	
For beginning with an indication that the acceleration is the derivative of velocity	1 point
$a(t) = \frac{dv}{dt}$	
For correctly evaluating the derivative	1 point
$a(t) = \frac{d}{dt}(-0.3t^2 + 7t) = -0.6t + 7$	
(e) 3 points	
For an indication that the displacement is the integral of the velocity	1 point
$\Delta x = \int_{t_0}^t v(t) dt$	
For attempting to integrate with appropriate limits or constant of integration	1 point
$\Delta x = \int_0^t (-0.3t^2 + 7t) dt$	
For the correct answer	1 point
$x(t) = -0.1t^3 + 3.5t^2$	
(f) 3 points	
For recognizing that the distance is the change in position	1 point
$\Delta x = x(8) - x(2)$	
For using the expression from part (e) to calculate $x(2)$ and $x(8)$	1 point
$x(2) = 13.2 \text{ m}$	
$x(8) = 172.8 \text{ m}$	
For the correct answer with units	1 point
$\Delta x = 159.6 \text{ m}$	
<i>Alternate solution</i>	<i>Alternate points</i>
<i>For using the integral of the speed and recognizing that the distance is the change in position</i>	<i>1 point</i>
<i>For using the correct limits or constant of integration</i>	<i>1 point</i>
$\Delta x = \int_2^8 (-0.3t^2 + 7t) dt$	
$\Delta x = \left. \frac{-0.3t^3}{3} + \frac{7t^2}{2} \right _2^8 = -51.2 + 224 - (-0.8 + 14)$	
<i>For the correct answer with units</i>	<i>1 point</i>
$\Delta x = 159.6 \text{ m}$	

**AP[®] PHYSICS C: MECHANICS
2015 SCORING GUIDELINES**

Question 2

15 points total

**Distribution
of points**

(a) 3 points

For a correct application of conservation of energy

1 point

$$K_1 + U_1 = K_2 + U_2$$

$$mgH = \frac{1}{2}mv^2 + mgy_P$$

For relating y_P to H and θ

1 point

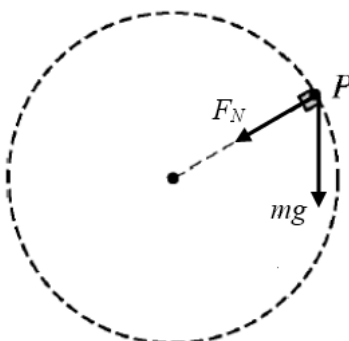
$$y_P = \frac{H}{2} + \frac{H}{2}\sin\theta$$

For a correct final expression for v , which does not have to be simplified

1 point

$$v = \sqrt{gH(1 - \sin\theta)}$$

(b) 2 points



For correctly drawing and labeling a vector for the weight of the block

1 point

For correctly drawing and labeling a vector for the normal force on the block

1 point

One earned point is deducted (up to a maximum of two points) for each extraneous force drawn

(c) 5 points

For applying Newton's second law with more than one force

1 point

For setting the sum of the forces in the radial direction equal to mv^2/R

1 point

$$F_N + w_{\text{radial}} = \frac{mv^2}{R}$$

For using the correct radial component of the weight

1 point

$$w_{\text{radial}} = mg\sin\theta$$

For correctly substituting for the speed and radius from part (a)

1 point

$$F_N + mg\sin\theta = \frac{mgH(1 - \sin\theta)}{H/2}$$

For an answer consistent with part (a)

1 point

$$F_N = mg(2 - 3\sin\theta)$$

AP[®] PHYSICS C: MECHANICS
2015 SCORING GUIDELINES

Question 2 (continued)

	Distribution of points
(d) 2 points	
For setting the normal force equal to zero (explicitly or implicitly)	1 point
$0 = mg(2 - 3\sin\theta)$	
$3\sin\theta = 2$	
$\theta = \sin^{-1}(2/3)$	
For an answer consistent with part (c)	1 point
$\theta = 41.8^\circ$	
(e) 3 points	
For correctly selecting $\theta_n < \theta_L$	1 point
For indicating that the sphere will be moving slower (or has less kinetic energy) when it loses contact	1 point
For indicating that some of the initial potential energy now goes into rotational kinetic energy	1 point
Example: Since some of the energy must go into the rotational energy of the sphere, the sphere will have slower translational speed at every point on the track. Since the sphere moves slower than the block, the sphere will lose contact with the loop at a lower height and a smaller angle.	

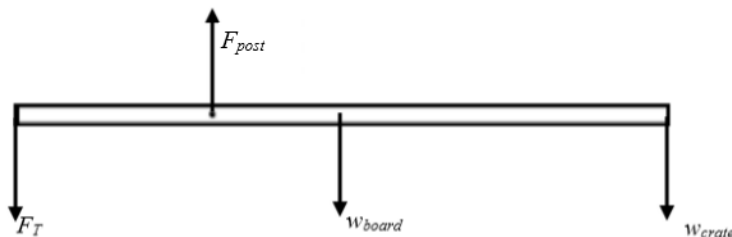
AP[®] PHYSICS C: MECHANICS 2015 SCORING GUIDELINES

Question 3

15 points total

**Distribution
of points**

(a) 3 points



For correctly drawing and labeling the three downward forces: tension, weight of the board, weight of the crate

1 point

For correctly drawing and labeling the upward force exerted on the board by the post

1 point

For drawing all of the forces at their respective points of action

1 point

One earned point is deducted (up to a maximum of two points) for each extraneous force drawn

(b) 4 points

For summing the torques about any labeled point and setting that sum equal to zero

1 point

Example using left end as pivot point:

$$\tau_{\text{cw}} - \tau_{\text{ccw}} = 0$$

$$\tau_{w\text{-board}} + \tau_{w\text{-crate}} = \tau_{\text{post}}$$

For substituting correct radii for all torques

1 point

$$(200 \text{ N})(1.2 \text{ m}) + (400 \text{ N})(2.4 \text{ m}) = (F_{\text{post}})(0.8 \text{ m})$$

For correctly calculating either the tension or the post force

1 point

$$F_{\text{post}} = 1500 \text{ N}$$

For a correct method for determining the remaining force and a consistent answer

1 point

$$F_{\text{up}} = F_{\text{down}}$$

$$F_{\text{post}} = w_{\text{board}} + w_{\text{crate}} + F_T$$

$$F_T = 1500 \text{ N} - 200 \text{ N} - 400 \text{ N}$$

$$F_T = 900 \text{ N}$$

AP[®] PHYSICS C: MECHANICS
2015 SCORING GUIDELINES

Question 3 (continued)

**Distribution
of points**

(c) 3 points

For correctly calculating the rotational inertia of the board about P

1 point

$$I_{\text{board}-P} = Md^2 + I_{\text{cm}} = Md^2 + \frac{1}{12}ML^2$$

$$I_{\text{board}-P} = \left(\frac{200 \text{ N}}{9.8 \text{ m/s}^2} \right) (0.4 \text{ m})^2 + \frac{1}{12} \left(\frac{200 \text{ N}}{9.8 \text{ m/s}^2} \right) (2.4 \text{ m})^2 = 13.1 \text{ kg}\cdot\text{m}^2$$

(or $12.8 \text{ kg}\cdot\text{m}^2$ using $g = 10 \text{ m/s}^2$)

For correctly calculating the rotational inertia of the crate about P

1 point

$$I = m_{\text{crate}} r^2 = \left(\frac{400 \text{ N}}{9.8 \text{ m/s}^2} \right) (1.6)^2 = 104.5 \text{ kg}\cdot\text{m}^2$$

(or $102.4 \text{ kg}\cdot\text{m}^2$ using $g = 10 \text{ m/s}^2$)

For recognizing that the total rotational inertia of the system about P is the sum of the inertia of the board and the crate about P

1 point

$$\Sigma I_P = 117.6 \text{ kg}\cdot\text{m}^2 \quad (\text{or } 115.2 \text{ kg}\cdot\text{m}^2 \text{ using } g = 10 \text{ m/s}^2)$$

(d)

i. 2 points

For summing torques about P , setting the sum equal to $I\alpha$, and substituting appropriate values

1 point

$$\Sigma \tau_P = w_{\text{board}} d_{\text{board}} + w_{\text{crate}} d_{\text{crate}} = I\alpha$$

$$(200 \text{ N})(0.4 \text{ m}) + (400 \text{ N})(1.6 \text{ m}) = 115.2\alpha$$

For an answer consistent with part (c)

1 point

$$\alpha = 6.13 \text{ rad/s}^2 \quad (\text{or } 6.25 \text{ rad/s}^2 \text{ using } g = 10 \text{ m/s}^2)$$

ii. 2 points

For a correct relationship between angular and linear acceleration

1 point

$$a = r\alpha$$

$$a = (0.80 \text{ m})(6.13 \text{ rad/s}^2)$$

For an answer consistent with part (d)i.

1 point

$$a = 4.9 \text{ m/s}^2 \quad (\text{or } 5.0 \text{ m/s}^2 \text{ using } g = 10 \text{ m/s}^2)$$

Units 1 point

For correct units on all answers

1 point